Experiment Name: 8-bit Addition using Assembly Language

1. Objective

The objective of this experiment is to perform 8-bit addition using Assembly Language in the 8086 microprocessor. This experiment helps to understand how arithmetic operations are performed using registers and instructions.

2. Apparatus / Tools

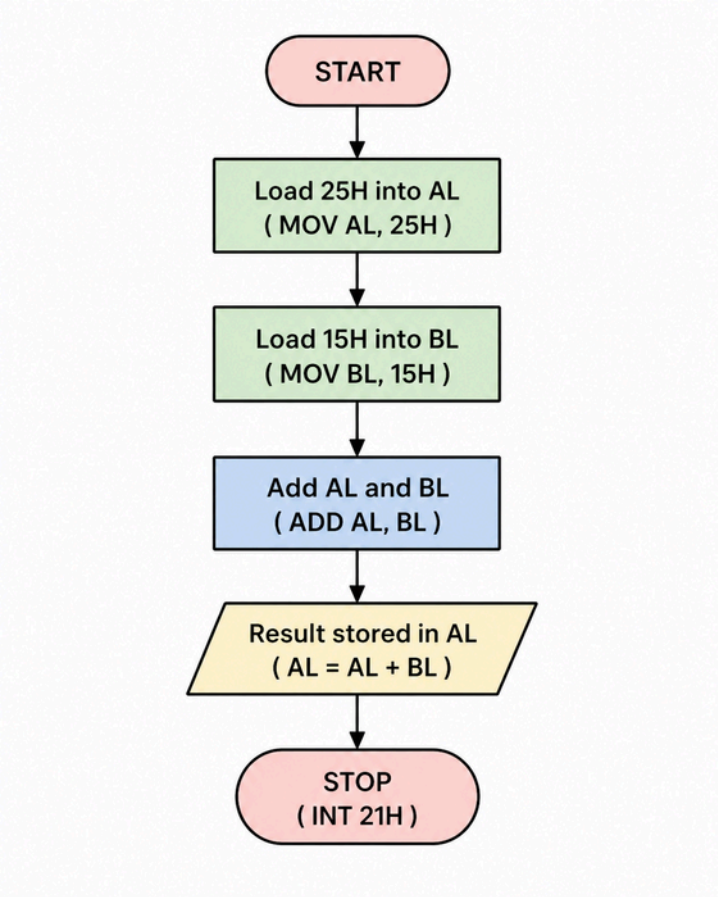
- Personal Computer (PC)
- EMU8086 Emulator Software

3. Working Procedure

Steps:

1. The first number is loaded into the AL register.
2. The second number is loaded into the BL register.
3. The ADD instruction is used to perform addition.
4. The result is stored in the AL register.

Flowchart:

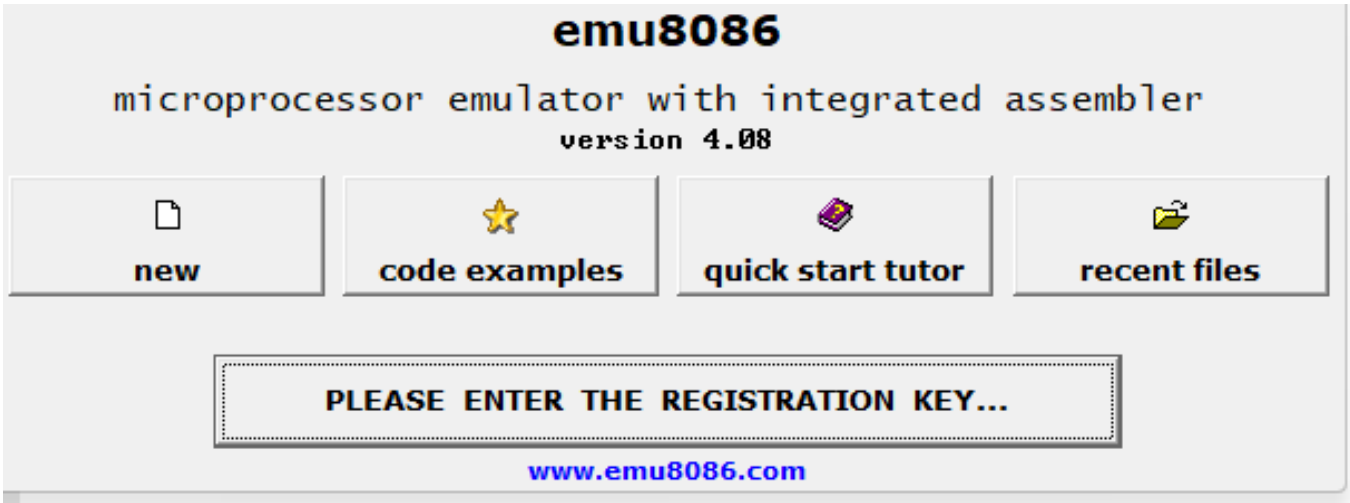


4. Program :

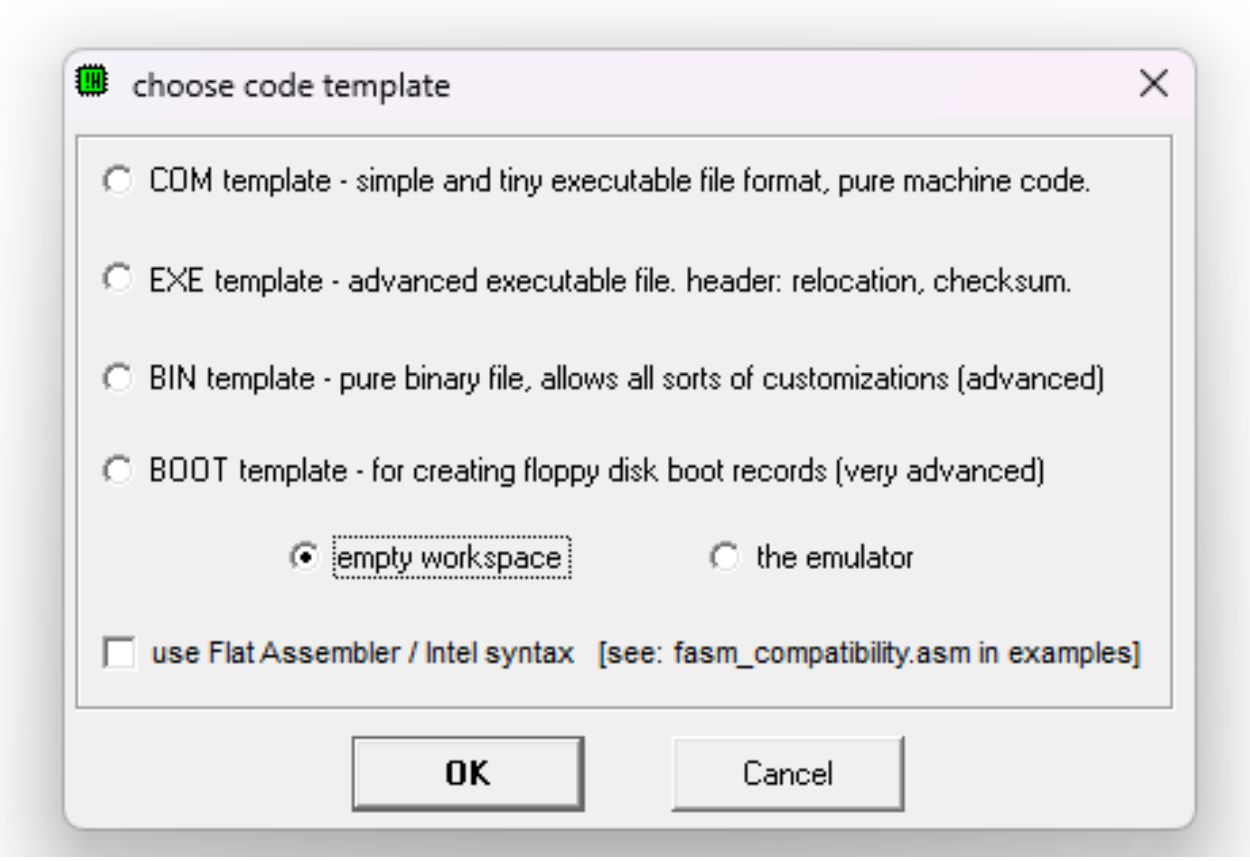
```
.MODEL SMALL
.STACK 100H
.DATA
.CODE
MAIN PROC
  MOV AL,25H
  MOV BL,15H
  ADD AL,BL
  INT 21H
  MAIN ENDP
END MAIN
```

5.Operation Step

S**tep 1**: Open EMU8086 and Click on “New”
First, open the EMU8086 emulator. After opening, click on the “New” option to start a new project.

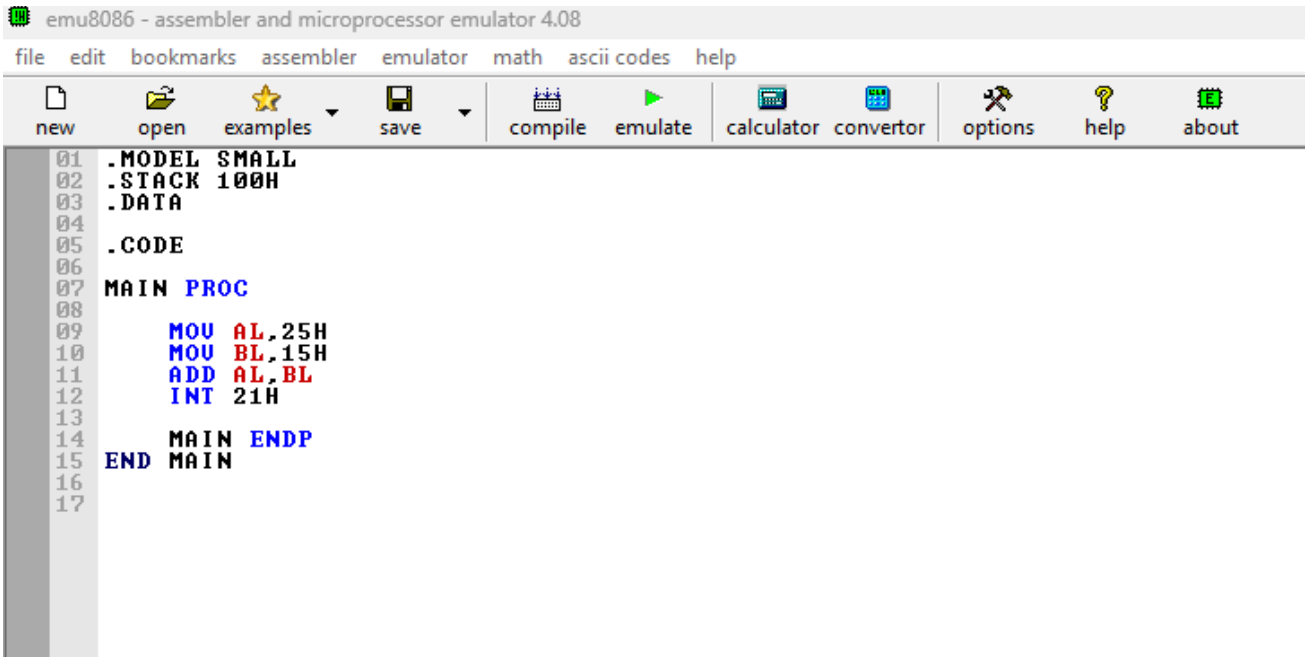


Step 2: Select “Empty Workspace”
After clicking on “New”, a new window will appear. From there, select “Empty Workspace” and click OK to proceed.



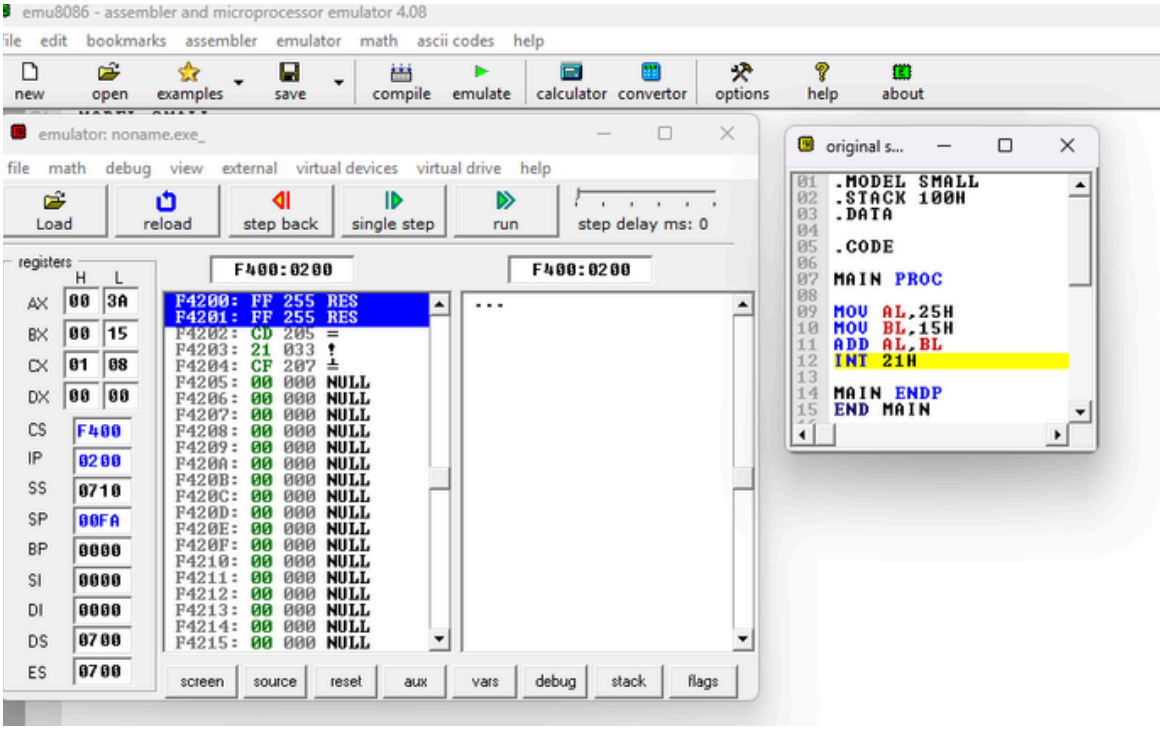
Step 3: Write the Program Code

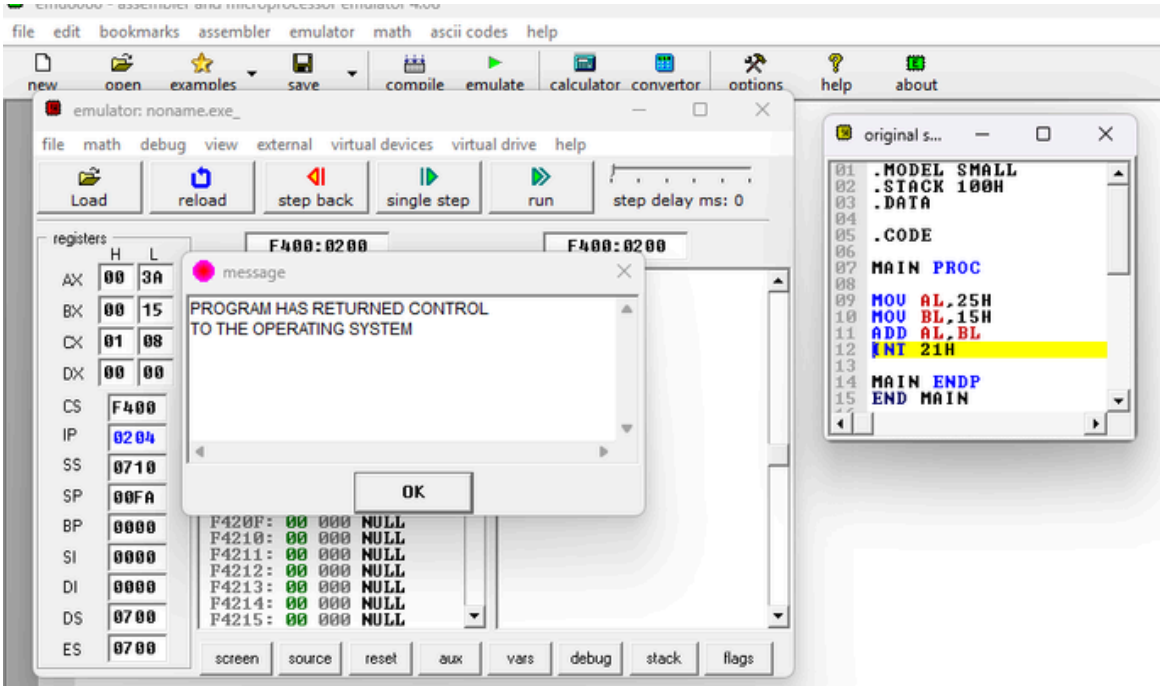
After opening the workspace, write the assembly language program in the editor window.



5.Operation Result

Step 1 :Click on the Run/Execute option to run the program. The result of the addition will be shown in the register (AL).





6.Comment

The program executed successfully, and the 8-bit addition was correctly performed with the result stored in the AL register.